Oral Surgery
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Complications of local anesthesia

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Complications of local anesthesia

- The widespread use of local anesthesia in dental practice today is in itself attribute to both the effectiveness and safety of the method. Nevertheless difficulties and complications occasionally occur and it is essential that the dental surgeon should know how to minimize their incidence. However he should never forget that any one of the many thousands of injections he give may cause complication or even a dangerous reaction and must take steps to ensure that he has both knowledge and means to diagnose and deal effectively with such a situation.

- A number of complications are associated with the administration of L.A. and those complications can be classified as local and systemic.

Local complications of local anesthesia include:

1. Pain and burning on injection
2. Failure to obtain anesthesia
3. Prolong impairment of sensation
4. Needle breakage
5. Facial paralysis
6. Ophthalmologic complications
7. Trismus
8. Soft tissue injury
9. Hematoma
10. Infection

1)Pain and burning on injection

- Pain on injection increases patient anxiety and may lead to sudden unexpected movement which may lead to increase risks of needle breakage.

Causes

1. Careless injection technique: an incorrect injection site can lead to an intramuscular or intraneural injection. When the needle touches a nerve; the patient feels a sudden electrical shock in the distal area of the nerve
2. Rapid deposition of the local anesthetic solution may cause tissue damage and pain.
3. PH of the solution being deposited into the soft tissue: the pH of local anesthesia as prepared for injection is 5, whereas that of solution containing vasopressor is even more acidic.
4. Contamination of the L.A cartridge can result when they stored in alcohol or other sterilizing solutions leading to diffusion of these solutions into the cartridge

5.Temperature of the solution: warmer solution is more comfortable to the patient than the cold one.

6.Aggressive insertion of the needle can tear the soft tissue, blood vessel, nerve, and periosteum and cause more pain.

7.Pain on withdrawal: pain on withdrawal of the needle from the tissue can be produced by fishhook barbs on the tip, this barbs may be produced during the manufacturing process but it is more likely that they occur when the needle tip forcefully contact a hard surface such as bone.

Prevention

1- Follow the technique of local anesthesia properly.

2-Use sterile local anesthetic solution which stored in correct temperature.

3- Inject local anesthesia slowly.

2)Failure to obtain anesthesia

- Although the incidence of this difficulty tend to decrease as the experience of the operator increases but it is still probably the most common problem seen during the use of local anesthesia. This problem is most common with block technique especially in the lower jaw the possible reasons behind this can be classified as:

Causes

- 1) Anatomical reasons: which include

a)Accessory nerve supply.

b)Abnormal course of the nerve.

c)Variation in the foramen location.

- 2)Pathological reasons are:

a) Trismus (limited mouth opening): in these it is impossible to use conventional technique of inferior alveolar nerve block.

- Therefore, the closed mouth technique is useful. In these case there will be an increased possibility of failure anesthesia.

b)Infection and inflammation: If the pulp is inflamed, the low tissue PH may cause lack of effective anesthesia in that area. However, this does not explain failure in block anesthesia where the solution is injected away from the inflamed area. A possible cause is hyperalgesia; the inflammation makes the nerve more sensitive to pain.

- 3)Faulty technique: Common mistake are:-
- a-An inadequate amount of local anesthetic solution being deposited in close proximity to the nerve.
- b-Injection L.A solution in blood vessels
- c-Injection of L.A solution away from the nerve and mainly if anatomical variation is present.

- 4)Psychological reasons: The fear and anxiety can cause failure in local anesthesia, to enable successful anesthesia, relaxation of the patient is needed.
- 5)When using a solution after the expiration date recommended by the manufacture.

3)Prolong impairment of sensation

- which can be defined as altered sensation beyond the expected duration of anesthesia or it is the prolonged loss of sensation. It is common in dental practice.
- When anesthesia persists for days, weeks, or months, there is an increased potential for self-inflicted injury, biting, thermal or chemical trauma which can occur without the patient awareness.

Causes

1. Trauma to any nerve during injection may lead to paresthesia. Patients report the sensation of an electric shock throughout the distribution of the involved nerve.
2. Injection of anesthetic solution that is contaminated with a neurotoxic substance such as alcohol or other sterilizing solution near a nerve produces irritation, resulting in edema and increased pressure in the region of the nerve leading to parasthesia.
3. Hemorrhage and infection in close proximity to a nerve.

Management:

Most paresthesia resolves within approximately 8 weeks without treatment. Reassure the patient that the condition is transient with strict follow up. If the damage to the nerve is severe, the paresthesia will be permanent.

4)Needle breakage

Breakage and retention of needles within the tissue has become an extremely rare occurrences because of the introduction of disposable needles however reports of needle breakage still appear.


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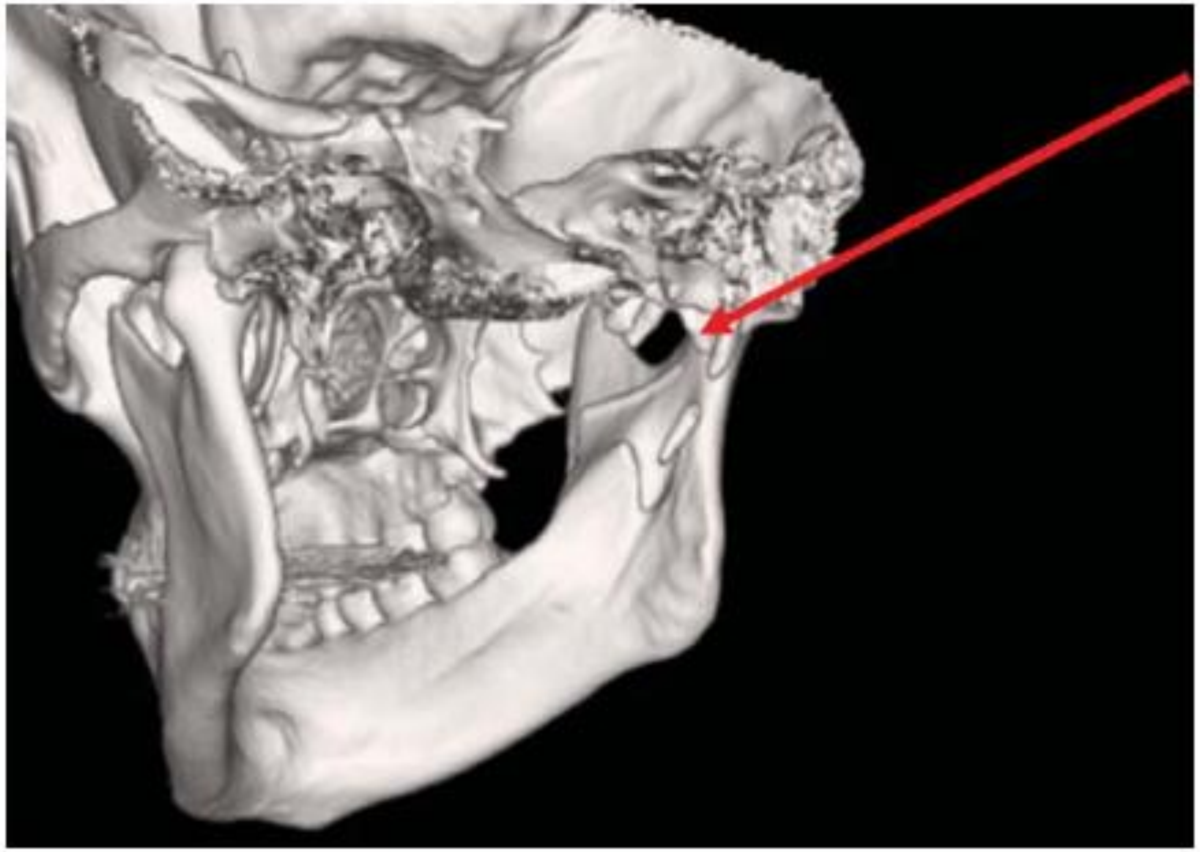

Causes

1. The primary cause of needle breakage is weakening of the dental needle by bending it before its insertion into the patients mouth.
2. The sudden unexpected movement by the patient as the needle penetrates muscle or contacts periosteum.
3. Change in the direction of the needle inside the tissue.
4. Too deep penetration as the needle goes up to its hub inside the tissue and might fracture at this point **the hub** is considered the most common point of needle fracture.

5.Using smaller needles as gauge (30) are far more likely to break than larger needles as gauge (25)

6.Re usage of the needle which its not accepted in dentistry now a day's, yet re usage may occur for the same patient during the same appointment when giving additional dosage of local anesthesia to the same patient, repeated injection cause fatigue of the needle structure and increases the risk of needle breakage

7.Aggressive insertion of the needle into the tissue.

8.Manufacturing defect.

Prevention

1. Using long needles for injection requiring penetration of significant depth of soft tissue.
2. Do not insert a needle into tissues to its hub unless it is essential for the success of the technique
3. Do not redirect a needle once it is inserted into tissues.
4. The dentist should check is any suspicion of inadequate product quality a new one should be used.

Management

- 1-stay calm and try to localize broken part in the tissue, then tell the patient what has happened and try to relax and comfort him.
- 2- stabilize the patient jaws in order that the needle stays in place ,if the patient move his jaw the tension from the muscle of masticatory system help the needle to penetrate the tissue.
- 3-if a portion of the needle is visible, grasp it firmly with instrument and remove it.
- 4- if you cannot remove the broken part yourself, refer the patient to oral and maxillofacial surgeon.

5)Transient Facial paralysis

- Paralysis of the facial muscles on one side is an uncommon complication of an inferior alveolar nerve block injection and may be either partial or complete depending upon which branch of the nerve are affected.

Cause

- This complication arises if the tip of the needle is inserted too far back and behind the ascending ramus.
- The solution is then deposited in the substance of the parotid gland where it anaesthetizes the branches of the facial nerve causing paralysis of the muscles they supply.

- ***Clinically*** the patient will immediately complain of transient paralysis of the muscle of the chin lower lip, upper lip, eye lid and inability to close the eye to raise the eye brow of the affected side.

- ***Management***
- Reassure the patient of the transient nature of the event and will last for few hours.
- Advice the patient to use an eye patch until motor function returns.
- If contact lenses are worn, they should be removed until the muscular movement retunes.

6-Ophthalmologic complications

- Amaurosis (loss of sight)
- Diplopia (double vision)
- Ophthalmoplegia (paralysis of all muscles responsible for eye movements)
- Ptosis (drooping of the upper eyelid).

Etiology:

Unknown may be attributed to the anesthetic solution reaching the orbit or nearby structures, causing ocular occlusion

Generally, these ophthalmologic complications have an immediate to short onset and disappear as the anesthesia subsides. mostly associated with intravascular injection when giving the inferior alveolar nerve block or the posterior superior alveolar nerve block. Therefore, it is recommended to do aspiration with these types of injections.

7)Trismus

- Trismus means limitation of mouth opening due to reduced mandibular mobility.

Trismus 3 days after a mandibular block injection.

Gradual forced opening with tongue blade therapy post-injection was required to resolve post-mandibular block trismus.

Causes

1. Trauma to the muscles the infra-temporal fossa, is the most common etiological factor in trismus associated with dental injection of local anesthetics.
2. Local anesthetic solution into which alcohol or cold sterilizing solutions have diffused produce irritation of tissues leading to trismus.
3. Hematoma in or around the muscles, the blood is slowly resorbed over approximately 2 weeks.
4. A low-grade infection after injection can also cause trismus.
5. Excessive volumes of local anesthetic solution deposited into a restricted area produce distention of tissues which may lead to post injection trismus

Prevention

1. Use sharp, sterile and disposable needle
2. Practice a traumatic insertion and injection technique.
3. Avoid repeat injections and multiple insertions into the same area through knowledge of anatomy and proper technique.
4. Use minimum effective volumes of local anesthetic solution.

Management

- Heat therapy, which consists of applying hot and moist towels to the affected area for approximately 20 min every hour or using warm saline rinse; a teaspoon of salt is added to a glass of warm water.

- Use analgesic and muscle relaxant.

- The patient is advised to initiate physiotherapy consisting of opening and closing the mouth as well as lateral excursions of the mandible, chewing gum is another means of providing lateral movement of the temporomandibular joint.

8)Soft-tissue injury

- **Soft tissue injury:** The soft tissue anesthesia lasts longer than pulpal anesthesia. Trauma to the anesthetized soft tissue can lead to swelling, pain and even infection.
- **Causes:** Self-inflicted trauma to the lips and tongue frequently occurs when the patient bites or chews these tissues while still anesthetized. Trauma occurs most frequently in younger children and in mentally retarded patients.

Lip sucking – this is the appearance of the lower lip of a 10-year-old male after bilateral mental nerve blocks for orthodontic bicuspid removal. Patient admitted to prolonged sucking of lip after the procedure while the lip was still anesthetized.

Three days status post inadvertent tongue biting after a mandibular block anesthesia.

Prevention

1. The local anesthetic of appropriate duration should be selected.
2. A cotton roll can be placed between the lip and the teeth if they are still anesthetized.
3. Warn the patient and his relative against eating, drinking hot fluid, and biting on the lips or tongue to test for anesthesia.

Management:

- Management is symptomatic:
- 1) Analgesics for pain.
- 2) Antibiotic if necessary.
- 3) Warm saline rinses to aid in decreasing any swelling present.
- 4) Use any lubricant to cover the lip lesion and minimize the irritation.

9)Hematoma

- A hematoma is a localized mass of collected blood outside the blood vessels that may become clinically noticeable following the injection. It is caused by penetration of the blood vessel with the needle during penetration or withdrawal of the needle in the tissue leading to bleeding inside the tissue and the formation of hematoma. The patient will notice development of swelling and discoloration of a bruise.

Hematoma that developed after bilateral mental nerve blocks.

The blood vessels most associated with hematoma are :

a) The pterygoid venous plexus
b) Posterior superior alveolar vessels
c) The inferior alveolar vessels in the pterygomandabular space.
d) The mental vessels
e) The infra-orbital vessels

- Prevention
- 1-learn anatomical landmarks and injection techniques.
- 2- Avoid relocating the needle to different sites inside the tissue.

- **Management**

- If it is visible immediately following injection, apply direct pressure if possible. Once bleeding has stopped, inform the patient of what was happened and reevaluate the possibilities of continuing the treatment. Instruct the patient to avoid application of heat over the area, prescribe analgesic and antibiotic if necessary.

- If it is invisible like in case of pterygomandibular space hematoma, the patient will come in the 2nd or 3rd day complaining of trismus, in this case, treat the case as trismus.

10)Infection

- Infection after the local anesthetic administration has be rare complication since the introduction of sterile disposable needle and g lass cartridges.
- ***Causes***
- 1-Contamination of the needle when touching the mucous membrane in the oral cavity before the administration of local anesthesia.
- 2-Improper technique in handling of the local anesthetic equipment.
- 3-Injecting of the solution into an area of infection which might transport bacteria into adjacent healthy tissues.

Prevention

- 1- Use sterile disposable needles.
- 2- Proper handle of the needle to avoid its contact with non-sterile surface.
- 3- Use cartridges only once and store it in their original container covered at all times.

Management:

- If an infection does occur the patient will complain of pain and trismus, immediate treatment consists of those procedures used to manage trismus. A course of antibiotic should be prescribed to the patient for 7 days.


THANK YOU